Fuzzy design RDD

Causal statistics for treatment models

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1 Research Question

> **Research Question**
>
> We want to know if receiving a loan (the treatment T) increases the chances to enroll at university (the outcome Y), using an RDD.

2 Fuzzy RDD Methodology

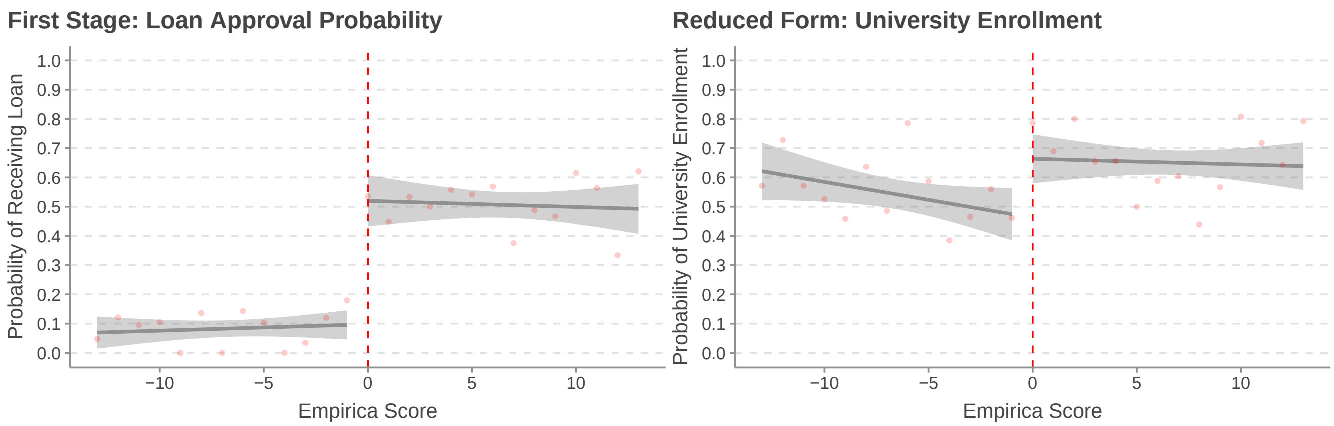
We implement a Fuzzy Regression Discontinuity Design (RDD) due to the following institutional setup:

- The Empirica score is used as the eligibility criterion for student loans.
- In theory, students with `Empirica > 0` should receive a loan.
- In practice, this rule is not strictly applied—some students below the threshold get loans, and some above do not—creating a "fuzzy" discontinuity.

Thus, we treat the threshold crossing as an instrument for actual loan receipt, using it in a Two-Stage Least Squares (2SLS) setup.

3 Visual Evidence

We begin by visually inspecting the discontinuity at the threshold.



The figure above illustrates the discontinuity at the Empirica score cutoff (0) for both loan approval and university enrollment:

- Left panel: There is a clear and sharp increase in the probability of receiving a loan just above the threshold. This confirms that the Empirica score functions effectively as an instrument, even if imperfectly enforced.

- Right panel: A visible jump in university enrollment at the same cutoff mirrors the pattern observed in the first stage, suggesting a potential causal link between loan receipt and enrollment decisions.

4 Regressions

4.1 First Stage Regression + Reduced Form Regression + 2SLS Estimation

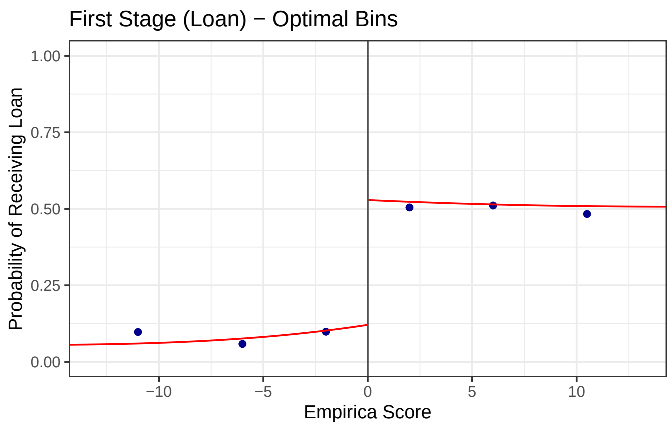
	FS (Loan ~ Threshold)	RF (Univ ~ Threshold)	2SLS (Univ ~ Loan)
(Intercept)	0.098 *	0.462 ***	0.415 ***
	(0.043)	(0.050)	(0.063)
discont_r	0.422 ***	0.202 **	
	(0.057)	(0.066)	
empirica	0.002	-0.012	-0.013
	(0.006)	(0.007)	(0.007)
empirica_r	-0.004	0.010	0.012
	(0.007)	(0.009)	(0.009)
loan			0.478 **
			(0.157)
N	876	876	876
R^2	0.199	0.016	0.013

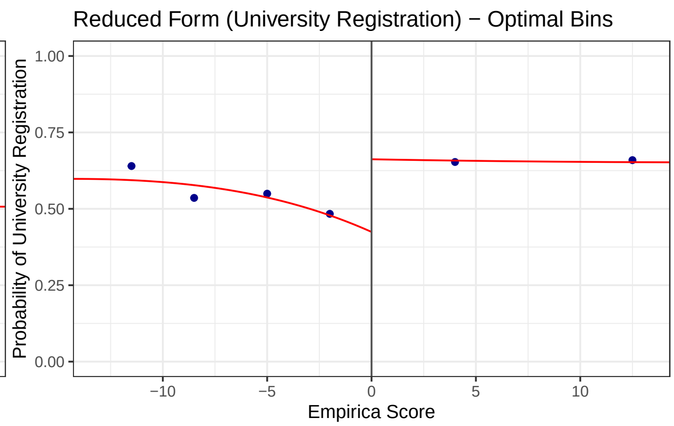
*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

4.2 Key Regression Results and Interpretation

Model	Key Coefficient	Estimate	Interpretation
First Stage Regression	`discont_r`	0.422	Students just above the Empirica threshold (repayment score > 0) are 42.2% more likely to receive a loan, confirming the instrument's strength.
Reduced Form Regression	`discont_r`	0.202	Students just above the Empirica threshold are 20.2% more likely to enroll in university, suggesting a positive reduced-form effect of loan eligibility.
2SLS Estimation (IV model)	`loan`	0.478	Receiving a loan increases the probability of university enrollment by 47.8%. This reflects the causal effect for compliers identified by the fuzzy RDD.

5 Confirm results with Robust Estimation (rdrobust)





Let's now estimate the effect of the loan on university registration using the rdrobust package.

```
Fuzzy RD estimates using local polynomial regression.

Number of Obs.                 5123
BW type                       mserd
Kernel                   Triangular
VCE method                       NN

Number of Obs.                 1189         3934
Eff. Number of Obs.             570          775
Order est. (p)                    1            1
Order bias  (q)                   2            2
BW est. (h)                  21.929       21.929
BW bias (b)                  38.072       38.072
rho (h/b)                     0.576        0.576
Unique Obs.                      64          208

First-stage estimates.

=============================================================================
        Method     Coef. Std. Err.         z     P>|z|      [ 95% C.I. ]
=============================================================================
  Conventional     0.405     0.047     8.556     0.000     [0.312 , 0.498]
        Robust         -         -     7.274     0.000     [0.292 , 0.507]
=============================================================================

Treatment effect estimates.

=============================================================================
        Method     Coef. Std. Err.         z     P>|z|      [ 95% C.I. ]
=============================================================================
  Conventional     0.468     0.142     3.299     0.001     [0.190 , 0.747]
        Robust         -         -     3.335     0.001     [0.226 , 0.870]
=============================================================================
```

Method	Estimate	Statistically Significant?	Method comparison
`rdrobust`	0.468	Yes (p = 0.001)	LATE estimated using robust local polynomial regression at the Empirica threshold[1].
2SLS	0.478	Yes (p = 0.0024)	LATE estimated using a global linear Two-Stage Least Squares (2SLS) specification.

Both methods yield consistent estimates of approximately 47%.

6 Conclusion

Fuzzy Regression Discontinuity Design analysis reveals a substantial causal effect of student loans on university enrollment among South African female students: obtaining a loan increases the likelihood of university enrollment by 47%.

[1]When using `rdrobust` for a fuzzy RDD, we need to specify the argument `fuzzy = data$loan`. Otherwise, the package will treat the RDD as a sharp design.